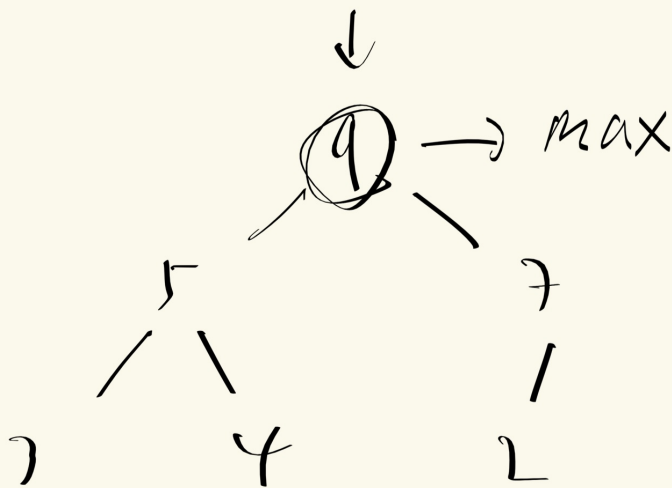
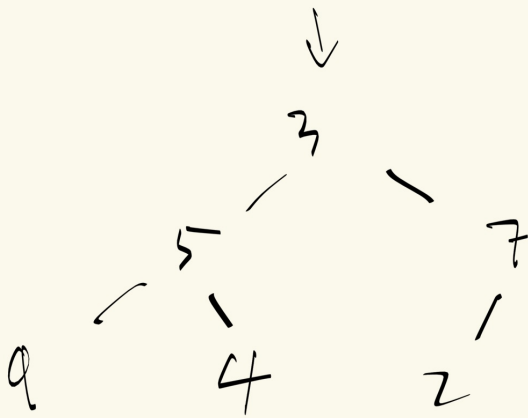


Heap Sort

- $O(n \log n)$
- Max heap
- Using Tree structure for array!

[3 5 7 9 4 2]



$\Rightarrow$ [9 5 7 3 4 2]
pop

[5 7 3 4 2 9]

↓

Do it again

```
def heap_sort(arr):
```

```
    n = len(arr)
```

```
    for i in range( $n // 2 - 1$ , -1, -1):
```

```
        heapify(n, i)  $\rightarrow$  Only parent nodes!
```

```
    for i in range(n-1, 0, -1):
```

```
        arr[i], arr[0] = arr[0], arr[i]
```

```
        heapify(i, 0)  $\rightarrow$  swap Root with  
                        the current last
```

```
def heapify(heap_size, idx):
```

```
    left = 2 * idx + 1 # idx's left child index
```

```
    right = 2 * idx + 2 # " right " "
```

```
    largest = idx
```

```
    if left < heap_size and arr[left] > arr[largest]:
```

```
        largest = left
```

```
    if right < heap_size and arr[right] > arr[largest]:
```

```
        largest = right
```

```
    if largest != idx:
```

```
        arr[idx], arr[largest] = arr[largest], arr[idx]
```

```
        heapify(heap_size, largest)
```